



Thermo-flow modeling of geometry evolution considering the velocity flow fields during pulsed-laser processing

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ABSTRACT

This paper evaluates the effectiveness of laser beam micromachining (LBMM) by analyzing the characteristics of the molten metal flow field during the development of micro-features on metallic surfaces. It establishes a foundation for understanding defects such as recast layer thickness, microcracks, and bulges arising from this flow and offers insights on controlling these issues. Effectively managing and optimizing these factors can help reduce the time and costs associated with the process. A transient coupled thermo-flow numerical model has been developed to study the hydrodynamic performance of bio-materials like titanium alloy (Ti-6Al-4V), which is widely used in biomedical and aerospace industries. The model considers the effect of the driving forces, viz., viscous force, thermocapillary force, and recoil pressure, which plays an essential role in geometry evolution. The mechanism of geometry evolution and molten metal flow behavior is analyzed as the pulse number increases, and the effect of pulse energy on geometry evolution is also predicted. The developed model predicted a depth of around 35–40 μm at the end of the 10th pulse. The results obtained from the developed model were compared with the published results, which verified the model's validity. Overall, the developed thermo-flow model and results obtained provided insights into the molten pool formation and geometry evolution during laser processing of Ti-6Al-4V.

1. Introduction

Over the previous two decades, LBMM operations have become very popular due to enhanced micro-sized feature fabrication demands such as micro-cutting, drilling, and engraving. Micro-sized features, including through holes in turbine blades and fuel nozzles, significantly enhance cooling and fuel efficiency. Meanwhile, micro-sized dimples improve surface properties and reduce drag across various applications [1,2]. The production of these micro-sized features necessitated large quantities, leading to a demand for rapid processing while maintaining stringent geometrical tolerances. Additionally, it was essential to minimize processing side effects, including the heat-affected zone, recast layer thickness, debris, and micro-cracks. These side effects are related to the laser processing parameters, viz., laser power, pulse duration, and pulse power density. Controlling and minimizing these side effects is still challenging for researchers [3].

The literature reported several unconventional micro-feature creation techniques, including electrochemical, jet, and electric discharge machining [4–6]. LBMM is preferred over these techniques due to its fast

processing, non-contact nature, and suitability for materials. A range of industrial lasers is available in the present scenario for micro-sized applications and can be categorized into ultrashort pulse regimes and long pulse regimes. In ultrashort pulse regimes, they emit laser pulses of light in the order of femtosecond to picosecond duration [7]. The high pulse power density due to minimum pulse duration and spot diameter during interaction directly vaporized the materials, leaving good surface quality and suppressing the heat-affected area [8,9]. Despite good surface quality, ultrafast laser micromachining has a low production rate and high equipment cost. In the case of long laser pulses, material removal occurs due to intense heating, melting, and molten material expulsion followed by vaporization. During the processing time, the molten pool sticks on the side surface, and the solidified edges of the geometry create metallurgical defects [10,11]. Sometimes, post-processing is required to remove these defects, which increases the production cost. Overall, it is observed that generating microscale features in metallic materials with minimum metallurgical defects while improving the surface properties, wear resistance, frictional coefficient, and wettability at higher production rates remains an open challenge for

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researchers.

Researchers reported several aspects of the experimental and numerical study on developing microscale features on metals and non-metals using long laser pulses. Micro-sized features were developed on chromium-coated piston ring surfaces to improve the tribological properties studied by Echilmaman in air, still water, and flowing water medium [12,13]. The authors observed that dimple depths ranged from approximately 15 to 50 μm , influenced by the pulse energy and nanosecond pulse duration used. Optimizing the size of these dimples is crucial for improving tribological performance in both dry and lubricated environments. Proper optimization of dimple size is essential for enhancing tribological performance in both dry and lubricated conditions. In a similar study, Rahman et al. [14] characterized the microcraters created on silicon surfaces using nanosecond pulse durations at various pulse power densities. The authors also examined the ablation yield with a single pulse and analyzed the geometric evolution of the microcraters. It was noted that the ejection of molten material deposited along the edges of the machined area is influenced by the laser processing parameters. Furthermore, the geometry evolution behavior of thermal barrier-coated materials was investigated computationally and experimentally by Wang et al. [15] to characterize the impacting effects of different laser parameters as well as delamination phenomenon in thermal barrier-coated Inconel 718 plates subjected to a millisecond pulsed laser. However, the model does not consider the impact of other factors, viz., phase change, molten pool formation, and other forces.

Literature reports that forming a molten pool during the laser processing of materials plays a crucial role in defect formation. Therefore, investigating the formation and flow behavior of the molten pool is essential for addressing these processing defects. Ai et al. [16,17] extensively studied the physics based behavior of the molten pool and flow field to reduce weld defects in laser welding of aluminum alloy T-joints with solder patches. Additionally, Ai et al. [18] critically examined the effects of molten pool flow behavior in another laser application, focusing on forming cladding layers in uphill wire laser additive manufacturing on inclined substrates. In LBMM, researchers have also studied different pulsing and beam-shaping techniques for reducing recast layer thickness [19,20]. The authors observed that a larger volume of liquid molten material is expelled from the surface due to the increased vaporization pressure zone cavity generated by the preheating sub-pulses.

Furthermore, an extra technique was implemented in LBMM to enhance quality and productivity [21,22]. It was found that adjusting the laser processing parameters can help minimize defects. However, factors such as recoil pressure, thermocapillary forces, and viscous forces that develop during molten pool formation significantly contribute to defect formation in laser-processed materials. Therefore, it is crucial to investigate these forces and their impacts within the LBMM process.

Literature reports few attempts at predicting temperature distribution and mechanism of geometry evolution during laser processing using ultrashort and long laser beam pulses numerically. Vora et al. [23] utilized numerical modeling to predict alumina material's surface topology and roughness during the laser process at increasing pulse rates with millisecond pulses. The results indicate that, as the pulse rate increases, the surface roughness also increases. The free surface evolution of a molten pool during steel laser spot welding was reported by Paul et al. [24], considering the Fe–S system (+ve surface tension coefficient), and found that surface topography is uneven during welding and solidification time. Some authors reported the impact of thermocapillary convection in a laser-generated molten pool; their findings showed that sulfur content-induced transitional thermo-capillary convection could alter the shape of the fusion zone and lead to molten pool oscillations [25–27]. Recently, Zhang et al. [28] examined the surface topography of textured substrate materials using a modified heat transfer and hydrodynamics model. Their research incorporated the level-set method and concentrated on a single pulse in scanning mode. It has been observed

that substantial research has been conducted on temperature distribution, geometry evolution, and molten pool formation using millisecond, nanosecond, and femtosecond pulses.

Based on the literature review, scant research was noted on predicting the molten flow behavior of metal and its geometry evolution during the LBMM process with microsecond pulses. This gap is particularly evident when accounting for the influences of viscous forces, recoil pressure, and surface tension. Consequently, developing a 2D transient numerical model to predict molten metal flow and geometry evolution behavior in the LBMM process under microsecond pulses is considered essential. The developed model in this work incorporates phase changes and the effects of viscous forces, recoil pressure, and surface tension, using titanium alloy (Ti-6Al-4V) as the substrate material, widely utilized in the biomedical and aerospace sectors. Moreover, it was thought essential to establish a foundation for understanding the defects such as recast layer thickness, microcracks, and bulges arising from the molten metal flow. Thus, this work was initiated to study the process parameters for their effective application to reduce the time and costs associated with the process. Also, the developed model considers more realistic phenomena such as the effect of the driving forces, viz., viscous force, thermocapillary force, and recoil pressure, which plays an essential role in geometry evolution. The following section elaborates the development of model, its validation, utilization of the model for important study on the physics-based study of the process in terms of molten pool formation and geometry evolution during laser processing of bio-material.

2. Development of the thermo-flow model

During the laser processing of micro-dimples, a narrow area is ablated when the incident thermal energy exceeds the material's threshold value, with the duration of the laser-material interaction varying from femtoseconds to milliseconds. In this brief interaction period, phase changes in the material occur due to the influence of viscous forces, surface tension, and Marangoni forces, which collectively shape the geometry of the dimples on the substrate. Fig. 1 illustrates the hydrodynamics of the molten pool and the driving forces at play during the laser processing.

In view of this, a thermo-flow model has been developed using the finite element method (FEM) to analyze heat transfer to the substrate material and the laminar fluid flow within the molten pool geometry.

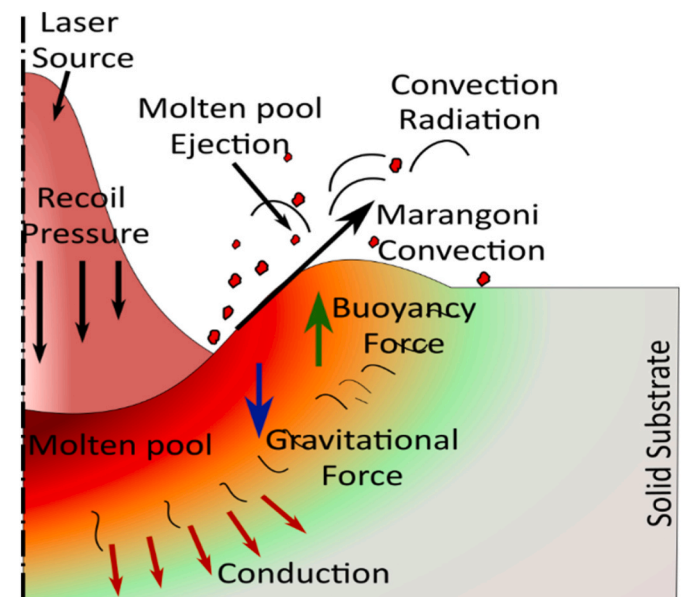


Fig. 1. Molten pool hydrodynamics and driving forces.

The schematic representation of the thermo-flow model is provided in Fig. 2. Various process parameters related to laser parameters and material properties (Fig. 3) have been considered. Using the 2D thermo-fluid model, we compute the temperature profile and velocity flow, which allows us to predict the evolution of the micro-dimple's geometry. Fig. 4 illustrates the process continuum used for the numerical simulation. While every effort has been made to enhance the accuracy of this model, some physical phenomena must be simplified or disregarded to streamline the LBMM procedure.

2.1. Assumptions

The assumptions considered for the development of the present model are as follows.

- Material is regarded as isotropic and homogeneous [34].
- The heat source is assumed to be of Gaussian shape.
- Inside the molten pool, the fluid flow is considered an incompressible laminar flow of Newtonian fluid [27,34].
- The absorptivity of the laser energy on the work surface is regarded as constant [29].
- The chemical reaction inside the molten pool is ignored [29].

2.2. Process continuum, governing equation, and boundary conditions

2.2.1. Process continuum

Fig. 4 displays a 2D model with boundary conditions for the thermo-flow model. The upper surface of the substrate material is irradiated with a laser heat source during the LBMM process. The heat source is assumed to be applied vertically and remains stationary.

A laser heat source is applied to boundary 2, with convection and surface radiation also considered at this boundary. The remaining boundaries, 1, 3, and 4, are kept insulated. The physical properties of Ti-6Al-4V alloy needed for the simulation are listed in Table 1.

2.2.2. Governing equation

The primary mechanism of material removal in LBMM is the thermal heating of substrate material. The laser heating causes phase changes

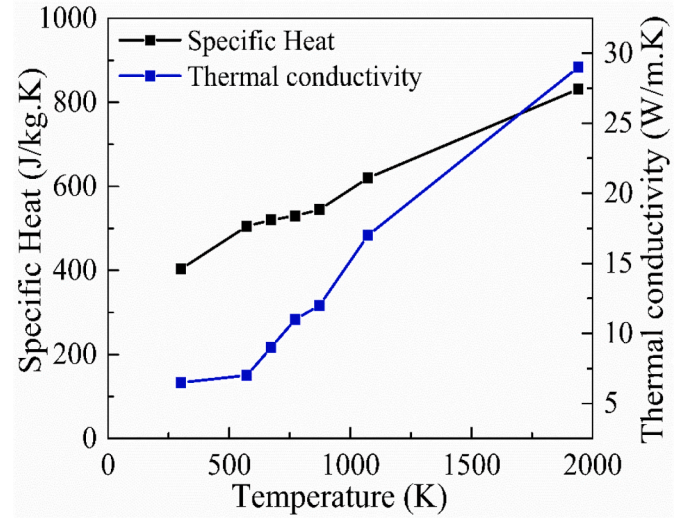


Fig. 3. Temperature-dependent properties of Ti-6Al-4V alloy [30,31].

from solid to liquid and then vaporization. Fourier's Law governs the heat transfer phenomenon, while the Navier–Stokes equations govern the velocity field inside the molten pool. These governing equations are shown below:

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p (\vec{u} \cdot \nabla T) = \nabla \cdot (k \nabla T) \quad (1)$$

$$\rho \left[\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} \right] = \nabla \cdot \left[-pI + \mu (\nabla \vec{u} + (\nabla \vec{u})^T) \right] + \vec{F} - \rho \vec{g} + \vec{F}_b \quad (2)$$

$$\rho (\nabla \cdot \vec{u}) = 0 \quad (3)$$

$$\vec{F} = -C_s \frac{(1-f_L)^2}{(f_L^3 + b)} \vec{u} \quad (4)$$

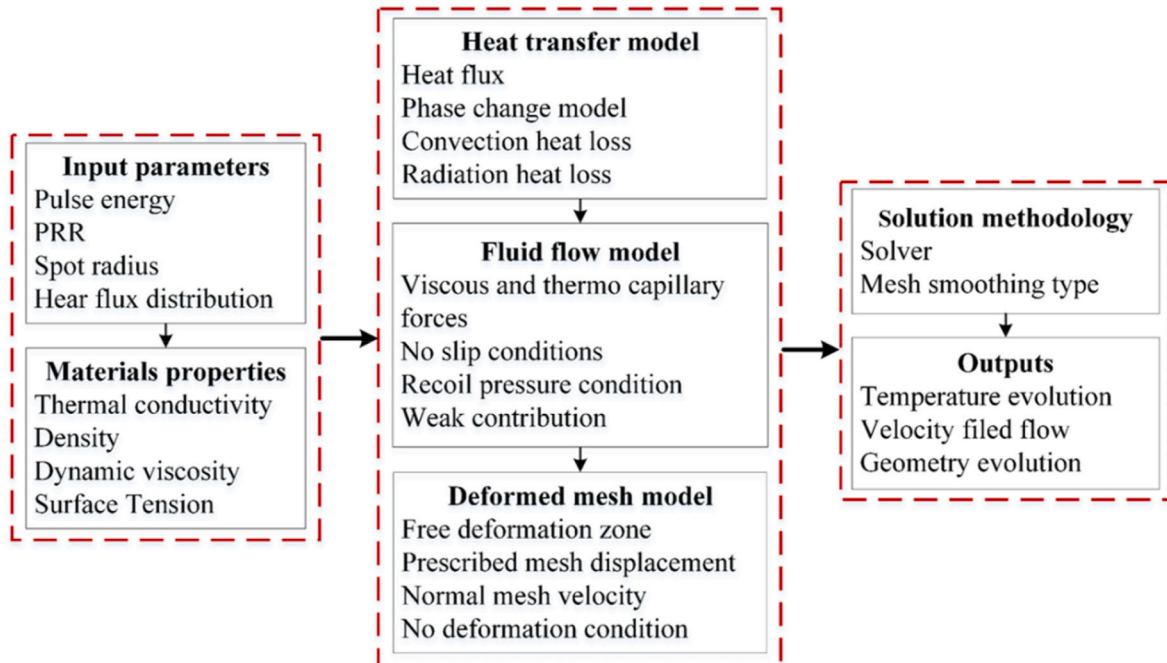


Fig. 2. Schematic representation of thermo-flow analysis.

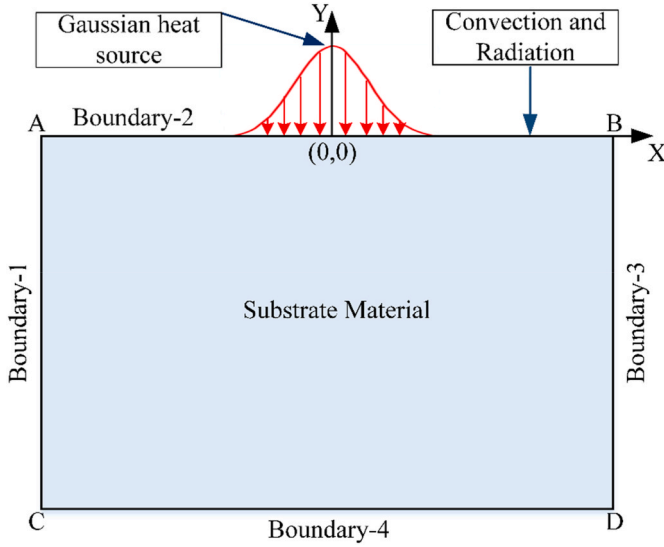


Fig. 4. Process continuum and boundary conditions.

Table 1
Physical properties of Ti-6Al-4V alloy [2,32,33].

S.No.	Properties	Symbol	Value
1.	Density	ρ	4430 (kg m ⁻³)
2.	Melting Point	T_{melt}	1941(K)
3.	Evaporation temperature	T_{vap}	3560 (K)
4.	Emissivity	ε	0.6
5.	Heat of fusion	H_{melt}	2.86×10^5 (J kg ⁻¹)
6.	Heat of vaporization	H_{vap}	9.83×10^6 (J kg ⁻¹)
7.	Stefan-Boltzmann constant	σ	5.670×10^{-8} (W.m ⁻² .K ⁻⁴)
8.	Ideal gas constant	R_{cons}	8.3140 (J K ⁻¹ .mol ⁻¹)
9.	Boltzmann's constant	β	1.38×10^{-23} (J K ⁻¹)
10.	Coefficient of thermal expansion	α	8×10^{-6} (K ⁻¹)
11.	Viscosity	μ	0.005 (Pa s)
12.	Atmospheric pressure	P_{atm}	101300 (N m ⁻²)

$$f_L = \begin{cases} 0 & \text{if } T \leq T_{\text{sol}} \\ \frac{T - T_{\text{sol}}}{T_{\text{liq}} - T_{\text{sol}}} & \text{if } T_{\text{sol}} < T \leq T_{\text{liq}} \\ 1 & \text{if } T > T_{\text{liq}} \end{cases} \quad (5)$$

where ρ is the density of the substrate material, p is pressure, $\vec{u}$ is the molten metal velocity during phase change, α is thermal expansion coefficient, k is the thermal conductivity, μ is the dynamic viscosity of the fluid, T_{melt} melting temperature, I is the identity matrix, $\vec{g}$ is the gravity acceleration, T_{liq} is liquidus temperature (slightly above melting point temperature), T_{sol} is solidus temperature (slightly below melting point temperature), and T is the absolute temperature. $\vec{F}$, the Darcy damping force is considered to dampen the velocity to zero when the temperature is lower than the melting temperature; this force induces isotropic permeability derived from the Kozeny-Carman Equation, which is also used to treat the mushy zone [35–38]. $\vec{F}_b$ is the buoyancy force. The thermo-fluidic boundary conditions applied to the computational domain are discussed in detail in the following sections.

2.2.3. Heat flux model and boundary conditions

A laser pulse energy is irradiated on the top surface of the substrate material. The input of the Gaussian heat flux can be expressed as [15]:

$$q_0(x, y, t) = \frac{2\alpha_0 E_p}{\tau_p \pi w_0^2} \exp\left(-2 \frac{x^2}{w_0^2}\right) h(t) \quad (6)$$

where $h(t)$ is the pulsing condition of the laser beam. It can be

mathematically expressed as

$$h(t) = \begin{cases} \text{on} & \text{if } t \leq \tau_p \\ \text{off} & \text{if } t > \tau_p \end{cases} \quad (7)$$

where q_0 is heat flux on the substrate surface, α_0 is the absorptivity; E_p is the pulse energy, w_0 is the spot radius of 10 μm , τ_p is pulse duration, and x is the distance from the beam center along the radial direction of the laser beam.

Heat flux boundary conditions are shown in Fig. 4. The top surface boundary-2 irradiated with the laser heat source is represented by equation (6). The same boundary-2 is considered to have a loss of heat due to convection and radiation, which can be defined as:

$$-\left(k_x \frac{\partial T}{\partial x}\right) - \left(k_y \frac{\partial T}{\partial y}\right) = h_t(T - T_{\text{amb}}) + \sigma \varepsilon (T^4 - T_{\text{amb}}^4) \quad (8)$$

where $k_x = k_y$, h_t , σ , and ε are the thermal conductivity, natural convective heat transfer coefficient, Stefan Boltzmann constant, and emissivity of the substrate material.

2.3. Fluid flow model and boundary conditions

In the LBMM process, the molten pool formation due to irradiated heat is subjected to three main driving forces: buoyancy, thermo-capillary, and recoil pressure, which induce complex flow patterns [2, 33]. These acting forces are shown in Fig. 1.

Buoyancy force (F_b) arises from temperature differences and mainly causes longitudinal flow in the molten pool. Mathematically, it can be expressed as [26,38]:

$$\vec{F}_b = \rho \vec{g} \alpha (T - T_{\text{melt}}) \quad (9)$$

where ρ is density, $\vec{g}$ is gravitational acceleration, α is the thermal expansion coefficient in liquid state, and T_{melt} is the melting point temperature.

Marangoni effect is the movement of the liquid driven by surface tension gradients that arise due to the temperature gradients; when the substrate material reaches its melting point during the machining process, a molten pool forms. The two driving forces are capillary forces, which usually act normally, and thermocapillary forces, which act tangentially on the deformation surfaces. The combined specific boundary conditions are shown in the following equations [38].

$$\sigma = (\nabla \cdot \vec{n}) \gamma \vec{n} - \nabla \gamma \quad (10)$$

where Capillary forces (surface tension) act in the normal direction.

$$\sigma_n = (\nabla \cdot \vec{n}) \gamma \vec{n} \quad (11)$$

The thermocapillary force acts tangentially, represented as [38].

$$\sigma_t = \nabla \gamma = \frac{\partial \gamma}{\partial T} \nabla T \cdot \vec{t} \quad (12)$$

where γ is the surface tension coefficient, $\frac{\partial \gamma}{\partial T}$ is the temperature derivative of surface tension, $\vec{n}$ and $\vec{t}$ is the normal and tangential vector of the free surface, respectively.

Another driving force, recoil pressure, is generated by the vaporization at the gas-liquid interface when the surface temperature exceeds the vaporization temperature of the substrate material. The following equation represents the recoil pressure applied on the top surface.

$$P_r = P_{\text{atm}} \times \exp\left[\frac{m H_{\text{vap}}}{R} \left(\frac{1}{T_{\text{vap}}} - \frac{1}{T}\right)\right] \quad (13)$$

where T_{vap} represents the evaporation temperature, T is the surface temperature, and H_{vap} is the latent heat of material vaporization and atmospheric pressure. Table 2 illustrates the boundary conditions for

Table 2

Applied heat transfer and fluid flow boundary conditions (Refer to Fig. 4).

S. No.	Boundary conditions	Boundary No.	Physics applied
1.	Heat Source	2	Laser Irradiation
2.	Surface Radiation	2	Radiation
3.	Convection heat transfer	2	Natural Convection
4.	Insulation	1, 3 and 4	No heat transfer
5.	Evaporative Heat Flux	2	Evaporation of particles
6.	Wall	1,3 and 4	No slip
7.	Outlet	2	Recoil Pressure
8.	Surface Tension	2	Marangoni effect
9.	Thermocapillary forces	2	Weak contribution

heat transfer and fluid flow modeling, and Table 3 shows the chemical composition of substrate material.

$$P_r = \begin{cases} P_{am}, & 0 \leq T < T_{vap} \\ \frac{1 + \beta_r}{2} \times P_{am}(T_s), & T \geq T_{vap} \end{cases} \quad (14)$$

2.4. Geometry evolution profile

Deformed mesh technology was adopted to predict the geometry's evolution during the modelling. Material removal was considered to occur when the surface temperature exceeded the vaporization temperature of the substrate material. During the simulation study, a convergence issue occurred, leading to an inverted mesh and an error that hindered the model from converging. The error is avoided using proper displacement constrained to the geometry in X and Y directions. Finally, for the movement of the mesh, a normal mesh velocity is designed based on the heat flux obtained from the heat transfer model. This normal velocity can be expressed mathematically as [40,41].

$$v_n = \frac{q_0}{H_{vap}\rho} \quad (15)$$

where q_0 is ablative heat flux, H_{vap} is latent heat of vaporization.

2.5. Mesh process continuum and solution methodology

A finite element method is used to solve the defined governing equations and their boundary conditions to compute the temperature profile, velocity profile, and pressure profile at the end of the pulse. The developed computational model and their boundary conditions were solved using COMSOL Multiphysics, a FEM solver in transient mode. The mesh process continuum of the geometric model is shown in Fig. 5(a) and (b). The entire substrate material is fine-meshed with an element size of 1 μm . The computational region comprises 24912 domain elements and 400 boundary elements of elements quality 0.9647. The full thermo-flow model was solved using the PARDISO solver. The computational time to solve the entire model was 5–6 h on core i7-32 GB RAM.

2.6. Estimation of energy for material processing

For the processing of the substrate material, sufficient energy will be needed to ablate the surface of the substrate material. For the selection of a micro pulse duration laser, the required energy (S) to melt and vaporize the material, the following equation can be used:

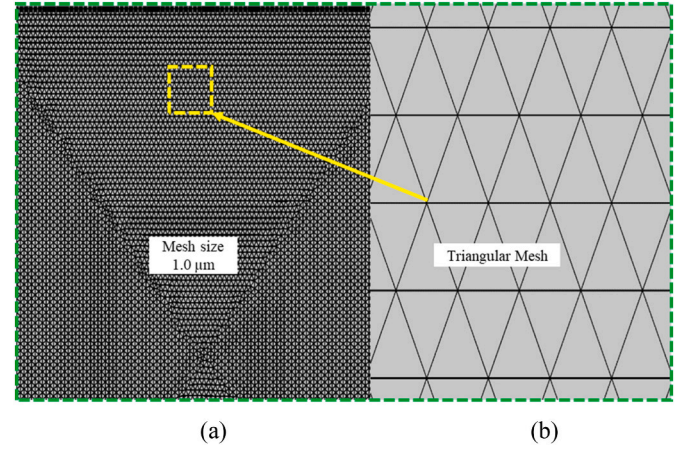
$$S = \rho(c_s(T_{melt} - T_{amb}) + H_{melt} + c_v(T_{vap} - T_{melt}) + H_{vap}) \quad (16)$$

where ρ is the material density, c_s , and c_v are specific heats, T_{melt} , T_{amb} ,

Table 3

Chemical composition of substrate material [39].

Ti-6Al-4V	Al	V	O	Fe	Cr	N	C	Ti
	5.5–6.75	3.5–4.5	0–0.2	0–0.4	0.021	0–0.5	0–0.1	balance

**Fig. 5.** (a) Mesh process continuum (b) zoomed view.

and T_{vap} are melting, ambient, and vaporization temperatures, respectively, and H_{melt} and H_{vap} are the latent heats of melting and vaporization, respectively. The following equations are used to get the pulse energy [42].

$$E_p = \frac{SV}{\alpha_0} \quad (17)$$

$$V = \frac{1}{3} \pi r^2 L_d \quad (18)$$

$$L_d = \sqrt{\alpha_t \tau} \quad (19)$$

where E_p is pulse energy, V is the volume of melting and vaporization, r is spot radius, L_d is thermal diffusion depth for given laser interaction time τ_p , and α_0 is absorptivity, whereas α_t thermal diffusivity.

Considering the energy at a pulse duration of 7.0 μs and spot radius of 10 μm , the pulse energy is approximately 102.4 μJ . However, the consideration of the other losses was ignored, so the energy varied at a different level, and the effect on surface geometry was predicted.

3. Results and discussion

The developed model simulates dimple formation with a single pulse and multi-pulse with process parameters reported in Table 4. Firstly, the model is validated with the published work of Li et al. [43] for a single pulse. Then, the model was extended to the present material for single-pulse and multi-pulse to predict the temperature profile, molten pool formation, velocity profile, effect of the recoil pressure, and geometry evolution.

Table 4

Laser process parameters.

S. No	Parameters	Value
1.	Pulse Duration	7 (μs), 10 (μs)
2.	Pulse	50 (kHz)
3.	Spot radius	10 (μm)
4.	Wavelength	1070 (nm)
5.	Pulse energy	325–400 (μJ)

3.1. Validation of model

The results predicted from our developed model are validated by comparing numerical and experimental results reported by Li et al. [43]. The exact process parameters were employed for the validation, as reported by Li et al. [43]. The validation of numerical and experimental results is reported in the following section.

3.1.1. Validation of temperature at a single pulse

Fig. 6 (a) and (b) represent the comparison of the temperature variation with time and in the radial direction, respectively, for a single pulse and at different pulse power densities of 29.920×10^4 (J/m²), 34.570×10^4 (J/m²), 47.750×10^4 (J/m²) and 117.770×10^4 (J/m²) with the published literature at the same process conditions as reported by Li et al. [43]. The dashed line represents the published model's data, and the continuous line represents our numerical data. The comparison revealed that the maximum temperature predicted by our developed numerical model agrees well with the results reported in the literature.

Fig. 6 (a) shows the maximum temperature below the melting point for a lower pulse power density of 29.920×10^4 (J/m²). Other pulse power densities of 34.570×10^4 (J/m²) and 47.750×10^4 (J/m²) are sufficient to exceed the melting point value but not sufficient to ablate the material from the surface. At the same time, the higher pulse power density of 117.770×10^4 (J/m²) is adequate to exceed the vaporization temperature of the substrate material.

Fig. 6 (b) shows the temperature variation along the radial direction at varying pulse power densities. It is observed that peak temperature increases with an increase in pulse power density. In the case of a higher pulse power density of 117×10^4 (J/m²), the peak temperature exceeds the vaporization temperature and leads to cavity formation. The reason for such variation may be that when the pulse power density increases, it induces a higher population inversion, which results in enhanced stimulated emission. The enhanced stimulated emission leads to an increased irradiation of photons on the substrate material.

3.1.2. Validation of crater profile at single pulse

Fig. 7 compares the predicted crater depth with the experimental results at a pulse power density of 117.770×10^4 (J/m²) reported by Li et al. [43]. It was found that at the given process parameter, only 117.77×10^4 (J/m²) pulse power density was sufficient to exceed the vaporization temperature of the substrate material. Therefore, it was considered for the crater depth validation. The results show a good agreement with the experimental data, although a 10 % error was observed. This

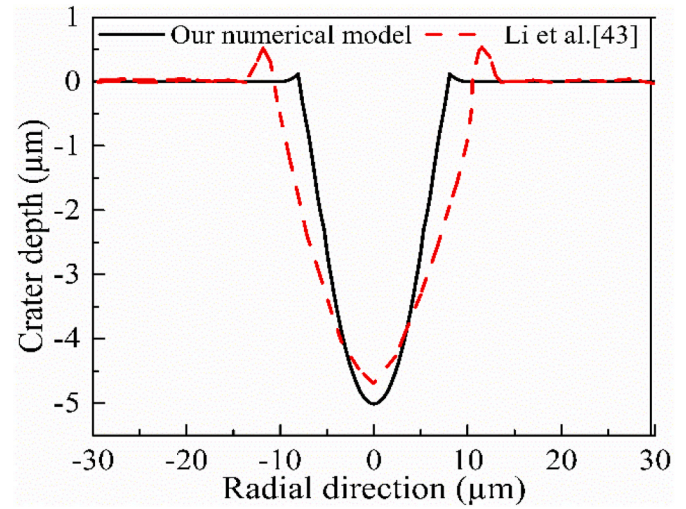


Fig. 7. Crater geometry at 117.77 (J/m²) pulse power density.

discrepancy may be attributed to the assumption of constant laser absorptivity during model development. Materials often exhibit different absorptivity at different temperatures or under varying conditions (e.g., surface roughness or coatings). Assuming constant absorptivity may overlook these factors, potentially leading to inaccurate results. Also, the occurrence of chemical reactions inside the molten pool can be the reason for the errors in the results. After confirming agreement with previously published work, the model was extended to the current material to investigate the effects of process parameters on temperature evolution, molten pool formation, and geometry evolution under single-pulse and multi-pulse conditions.

3.2. Temperature distribution and phase prediction on laser irradiation

Fig. 8 (a) and (b) represent the simulation results of the temperature evolution and isotherm contour at the initial stage of heating the material at a pulse duration of 7.0 µs. The heating and cooling times are 7.0 µs and 13 µs, respectively. When the highly concentrated laser heat source emits heat during the heating step, the material's surface temperatures rise above the vaporization point and then begin to vaporize. Due to the Gaussian nature of the laser intensity, it becomes higher at the center and decreases exponentially with distance from the center. It was

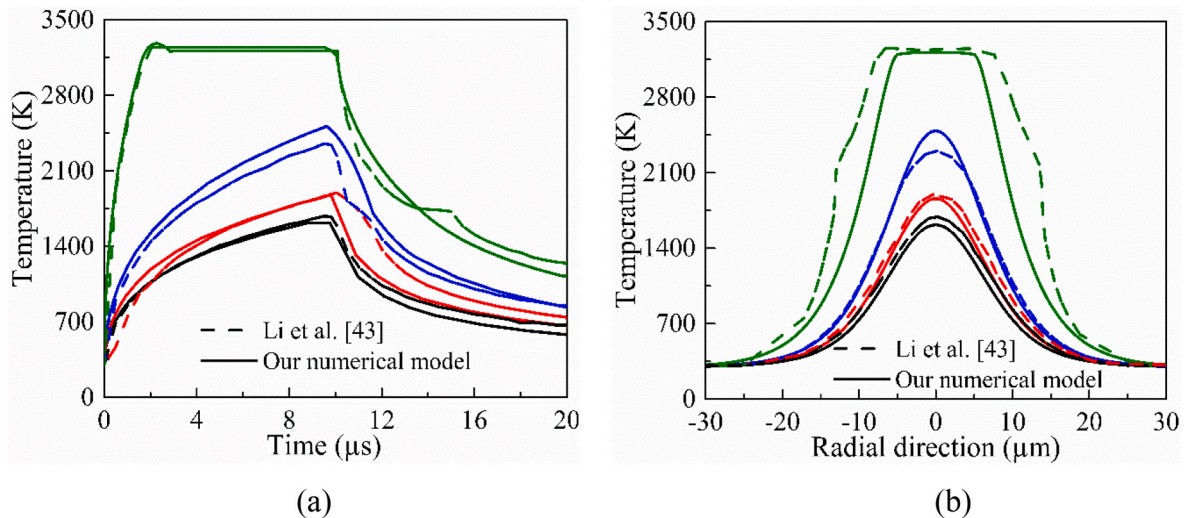


Fig. 6. Temperature profile at different pulse power densities with (a) time and (b) radial direction (1)at 29.920×10^4 (J/m²), (1)at 34.570×10^4 (J/m²), (1)at 47.750×10^4 (J/m²) and (1)at 117.770×10^4 (J/m²).

noticed that a small region goes into the vaporization state, followed by the melting. The remaining portion of the substrate material remains stable, represented in the temperature contour plot in Fig. 8 (b). It also shows the different phases of the material; the surface temperature of more than 3560 K is considered the vaporization phase, the liquid/vapor interface at a certain depth, and the material becomes in the liquid phase. Solidus temperature in the contour plot represents the interface of liquid and solid. In this region, material becomes liquid and solid, known as the mushy zone. The governing forces mainly act in this region and decide the shape of the geometry. The area below the solidus line has some defects, such as the heat-affected zone, as reported by the literature. The remaining portion remains unaffected as heat transfer is attenuated beyond a certain depth following Beer-Lambert's principle. Fig. 9 shows the temperature distribution over time for ten consecutive pulses. The temperature is predicted at the edge of the ablation region. The expected temperature shows the temperature change in the ablating region during the heating period while a decreased temperature during the cooling period. The rising temperature in successive pulses can be recorded due to the preheating nature of the surface.

Fig. 10 (a) and (b) represent the temperature variation along the depth and radial directions, respectively. The straight line in these figures represents the first pulse's material melting point and vaporization temperature. It shows that the first pulse was sufficient to ablate the material from the surface, as indicated in the red circle. After removal of the material from the surface, when the laser is off (cooling time), the surface temperature starts decreasing along the depth and radial direction, as seen in the figure after that, becomes constant.

3.3. Analysis of velocity field profile

The increase in the surface temperature creates a high molten pool. The temperature gradient and the resultant buoyancy effects influence the velocity of the molten material. It can be observed that maximum temperature occurs at the center of the focused laser, causing the higher molten pool to experience rapid expansion and movement. Also, variations in the molten metal's temperature cause variations in the metal's density, which produces flow into the molten pool due to buoyancy. The current model considers this effect by using the Boussinesq approximation. Fig. 11 shows the variation of velocity at different increasing pulse numbers. The velocity field goes up with the pulse numbers. The velocity profile shows a clockwise flow is developed, which transfers the "cold molten metal" from the bottom molten pool to the "top surface" and transfers the "hotter molten metal" from the top center surface to the edge of the molten pool. The inside legend represents the velocity field,

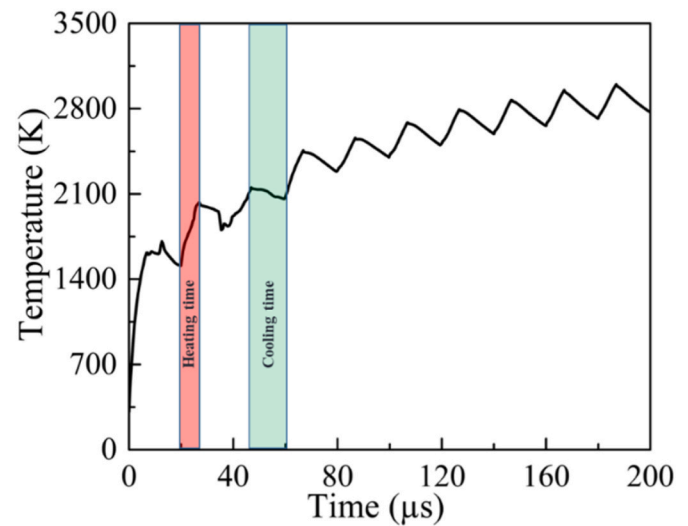


Fig. 9. Temperature evolution for multi-pulse at 7.0 μ s pulse duration and 50 kHz pulse repetition rate.

while the outer legend represents the temperature profile of the surface. The maximum velocity attained by the liquid metal is eight m/s at the end of the 10th pulse. It is also analyzed from Fig. 11 (c) – (e). The edges of the geometry show the downward movement of the flow, which will try to push back the molten material inside the cavity. Due to vaporization, the upward molten metal flow velocity is more important. In Fig. 11 (f), at the end of the pulse, the edges of the micro-dimples at solidification predict the accumulation of the recast layer thickness, which has been reported in experimental work [44]. Due to the increasing pulse number, more material evaporates, which causes an increment in aspect ratio, while fewer materials get remelted and resolidified on the edges. Moreover, the effect of surface tension was analyzed to verify the molten pool's flow characteristics. Fig. 12 (a) and (b) show the temperature coefficient of the surface tension and surface tension at the temperature distribution from the melting point to the vaporization point, respectively, since the effects of active elements are not considered in the current model. The surface tension declines with an increase in the surface temperature, as shown in Fig. 12 (b). Fig. 10 (b) shows that the surface temperature decreases along the radial direction, which may increase surface tension, causing the molten pool to move outward towards the edge side. The surface tension results were

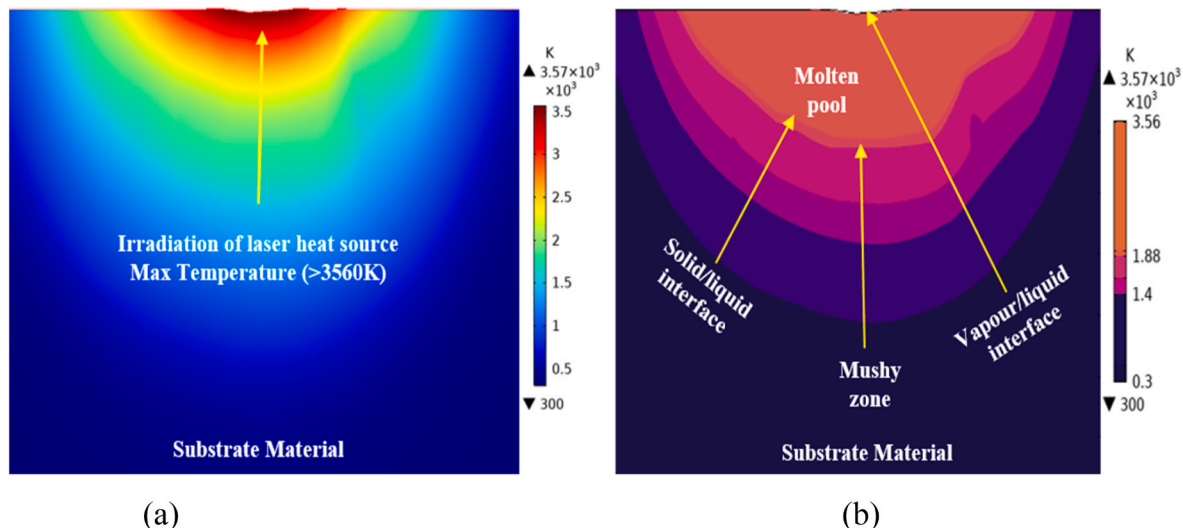


Fig. 8. (a) Distribution of temperature at 1.2 μ s (b) Isotherm contour of temperature distribution.

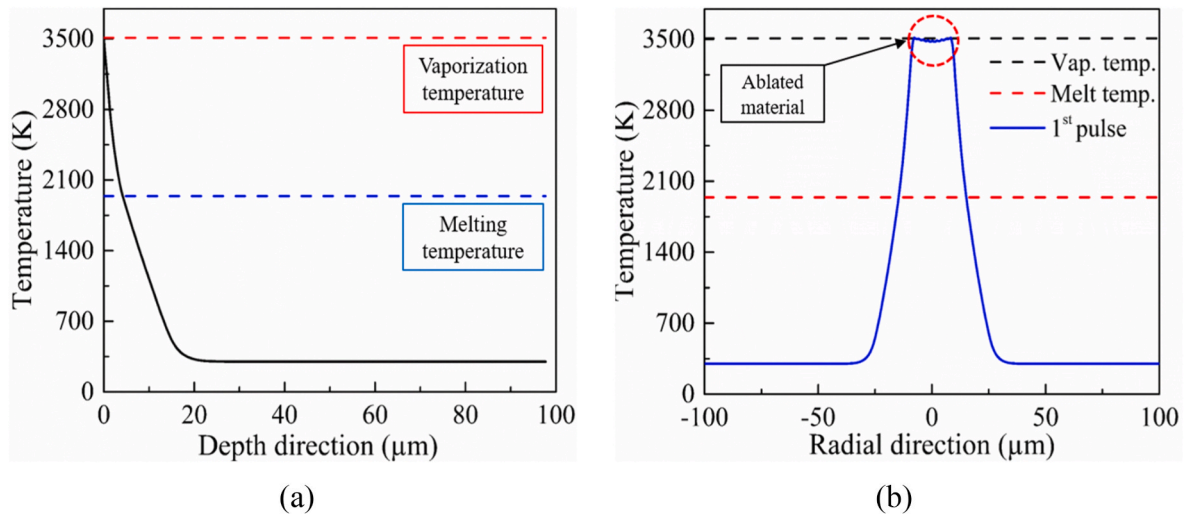


Fig. 10. Temperature distribution (a) along depth direction (b) along radial direction for the 1st pulse of heating.

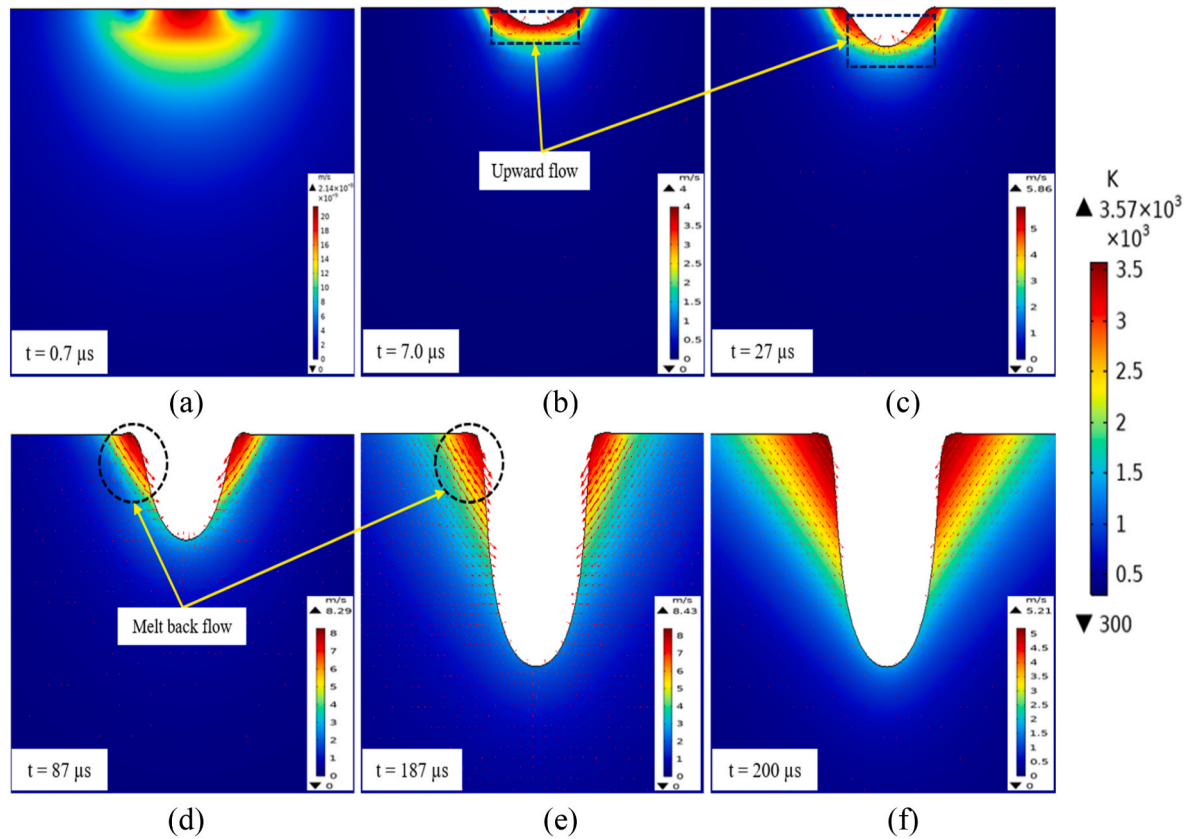


Fig. 11. Transient temperature field (Colour plot) and Velocity field (colored arrow plot) (a) $0.7 \mu\text{s}$ (b) at the end of 1st pulse (c) at the end of 2nd pulse (d) at the end of 5th pulse (e) at the end of the 10th pulse (f) at the cooling period.

also compared with the published literature by Li et al., who found similar effects during the investigation of stainless steel 304L alloy [43].

Fig. 13 (a) and (b) show the variation of the velocity field along the radial and depth direction. The velocity field was observed to become maximum at the center of the irradiating zone due to the maximum temperature produced at the center point. Further, it decreases along the edge due to the decreasing temperature and depth. The top velocity field was reported for the 10th pulse. After the end of the pulse (during the cooling period), its value decreases along the radial and depth directions due to the reduction of the temperature value in both directions.

Due to the effect of capillary and thermocapillary forces, the liquid at the free surface in the molten pool was pulled sideways at a high velocity. In contrast, the cold liquid near the molten pool glided to the center from its edge. Thus, it creates dynamic fluctuations (clockwise and anticlockwise vortices) in the molten pool. This molten pool behavior also has been reported by Chen et al. [45].

3.4. Study of dimple shape evolution at single-pulse laser and multi-pulse

The evolution of dimple geometry can be seen in Fig. 14 and 15. The

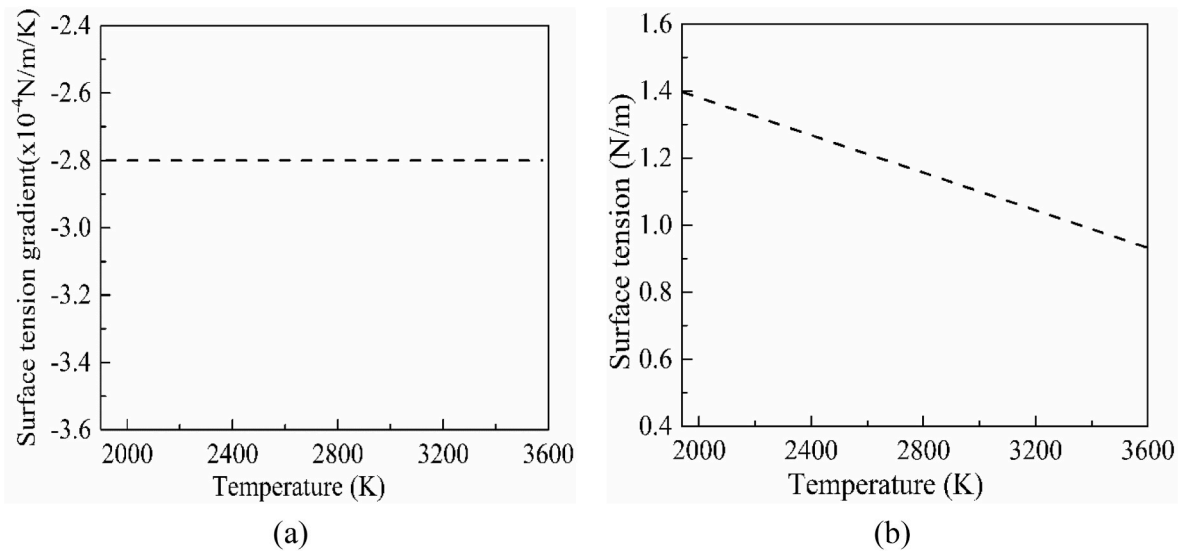


Fig. 12. Influence of surface tension gradient and surface tension.

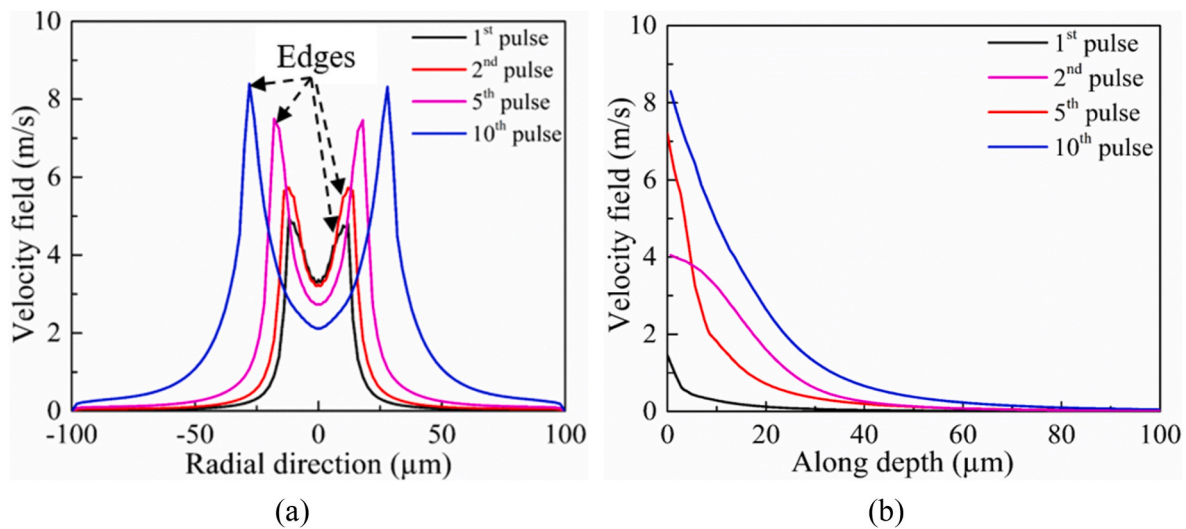


Fig. 13. Velocity field profile (a) along the radial direction (b) along the depth direction at various pulse numbers, at $7.0 \mu\text{s}$ pulse duration.

dimple depth increases from 1st pulse to the 10th pulse. The normal mesh movement of the elements is kept at the top surface. It was considered that the removal of material would occur when the surface temperature exceeded the vaporization temperature of the parent material. This normal mesh velocity is calculated based on the heat flux obtained from the heat transfer model [41]. Dimple depth increases with the increasing pulse number. As reported by the other literature, some edge bulges are predicted during the 4th pulse, which can be due to the more molten pool reflow at the edge. Due to increasing pulse number, the interaction time during heating rises, leading to more molten pool formation and the material evaporation time, causing more ablation. In particular, engineering applications of surface improvements in frictional, lubrication, and wear need deeper dimple sizes at minimized defects. The reported dimple depth at the end of the 10th pulse was around $35\text{--}40 \mu\text{m}$, which might be sufficient for surface improvement. It has also been reported that a good aspect ratio (depth/diameter ratio) gives better tribological performance due to a reduction in frictional and drag coefficient [46].

Fig. 15 also depicts the increasing depth in the line diagram. X and Y-direction show the radial distance and depth direction, respectively. From the 1st to 10th pulse number, the dimple depth and diameter

increased. Moreover, at the end of the 4th and 5th pulse numbers, some bulges can be seen near the edges of the dimples, which are due to the accumulation of the molten pool at the edges. Due to the short interaction time between the laser and material, the bulges' predicted size is significantly less than the millisecond pulses. It is also expected from the model that with an increase in the pulse number, the diameter of the dimple size increases due to the variation in the focal point moving along the ablation direction, which has also been validated through experimental work [47]. The reported work predicts the aspect ratio (depth/diameter) around 60–65 % could be attained.

3.5. Influence of pulse energy on geometry evolution

The influence of pulse energy on geometry evolution is predicted using this numerical simulation at the end of the 10th pulse number and 50 kHz pulse repetition rate for the pulse duration of $7.0 \mu\text{s}$ and $10 \mu\text{s}$, as shown in Fig. 16(a) and (b). It analyzed that by increasing the pulse energy from $325\text{--}400 \mu\text{J}$, the ablation depth increases in both cases for 7.0 and $10 \mu\text{s}$ pulse duration. The increasing pulse energy causes more energy interaction to the surface and more molten pool region formation. However, increasing pulse duration leads to higher ablation depth

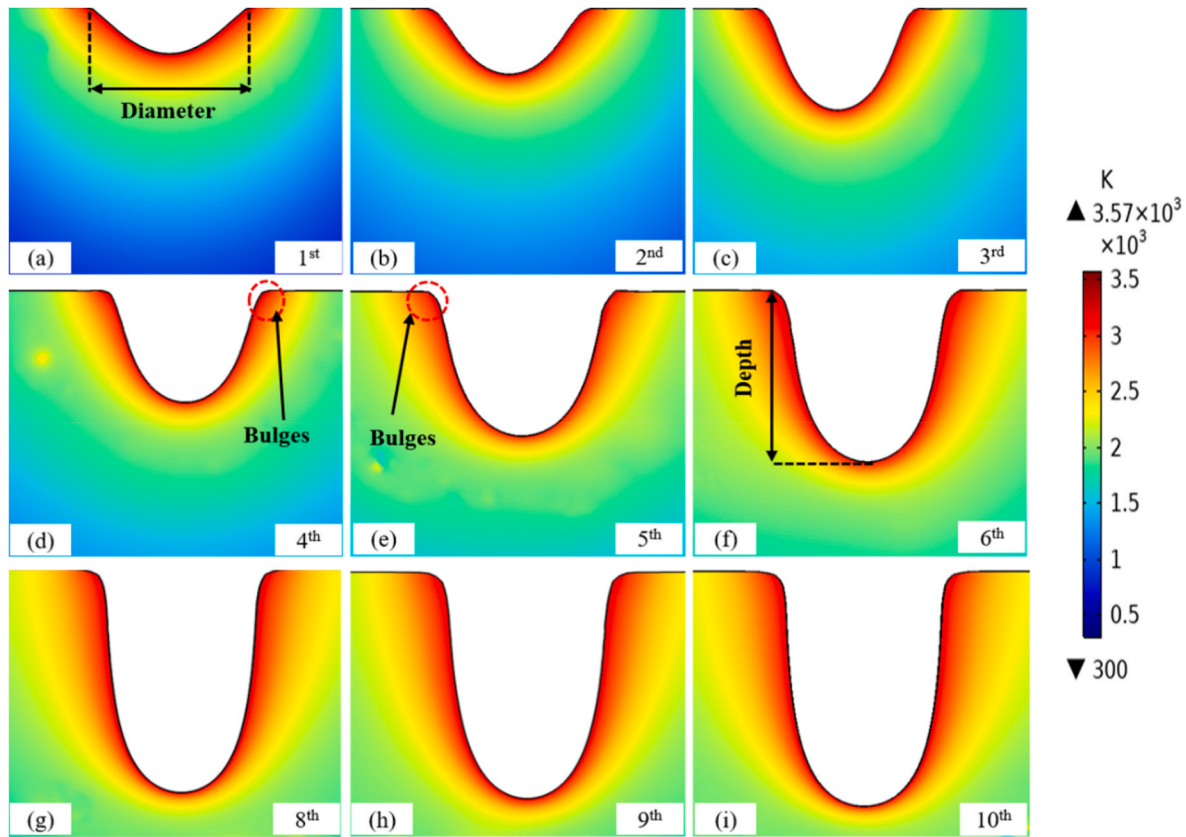


Fig. 14. Geometry evolution at different pulse no.

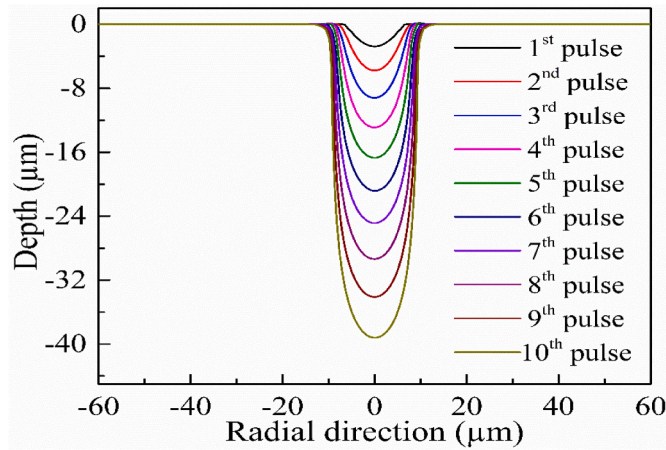


Fig. 15. Evolution of micro-dimples geometry from 1st pulse to 10th pulse.

than a lower pulse duration of $7.0 \mu\text{s}$ due to the more interaction time between the laser and surface. Also, increases in pulse duration decrease the cooling time, leading to a rise in surface temperature for the consecutive pulses and material removal from the surface.

4. Conclusions

This study presents the development of a transient 2D thermo-flow computational model for microsecond pulse for LBMM of bio-material titanium alloy (Ti-6Al-4V), focusing on molten pool dynamics, flow behavior, and geometry evolution. The model simulates transient molten metal flow patterns and geometry changes across ten laser pulses. The influence of processing parameters was analyzed using this

model, with predictions regarding the effects of pulse duration and energy on geometry evolution. The findings suggest that micro-dimples with an aspect ratio of 60–65 % can be reliably achieved within ten pulses. Key conclusions from this work include.

- Considering realistic parameters such as the effect of driving forces, viz., viscous force, thermocapillary force, and recoil pressure, made the developed model robust and helped in the accurate prediction of geometry evolution of dimples during laser processing of bio-materials.
- During the ablation of material, exceeding the vaporization temperature of the surface induces the recoil pressure, which helps remove the material.
- The shape of the molten pool increases as the pulse power density increases and decreases with cooling time. With an increasing number of pulses, bulges appear near the edges, and the formation of recast layer thickness is observed near the edges.
- The molten pool velocity increases as the temperature rises and becomes maximum at the heat irradiation point. It also increases with an increasing number of pulses. The predicted molten metal backward flow typically affects the quality of the product's final shape and the material properties. During the cooling phase, the molten pool solidifies, forming a recast layer and cracks over the surface.
- The relationship between laser processing parameters and the geometry of micro dimples indicated that the diameter and depth increase as pulse energy and pulse duration rise.

CRediT authorship contribution statement

Brijesh Kumar Singh: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sajan Kapil:** Supervision, Investigation, Conceptualization.

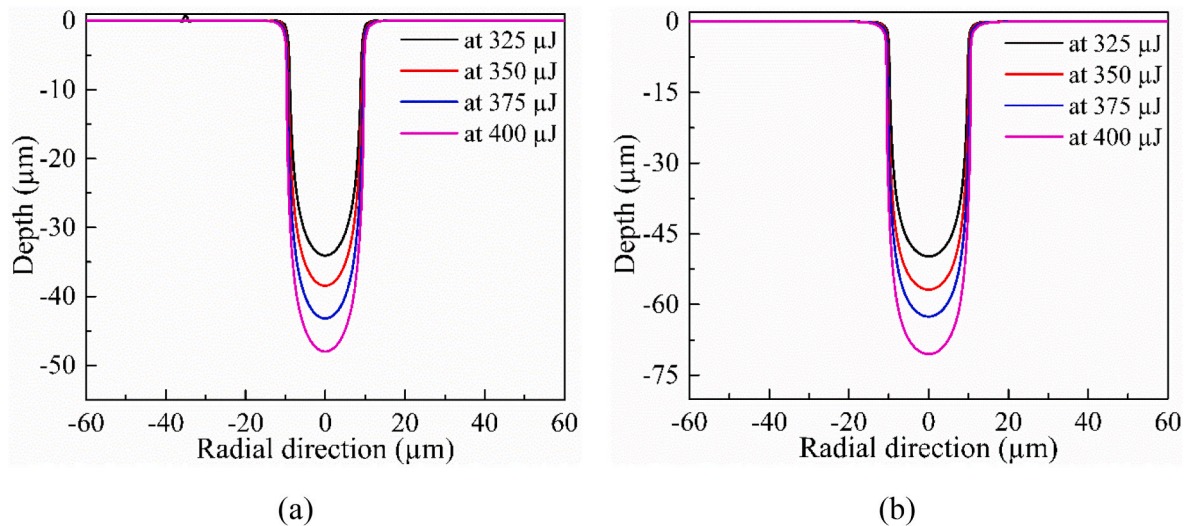


Fig. 16. Influence of pulse energy at (a) 7.0 μ s and (b) 10 μ s pulse duration.

Shrikrishna N. Joshi: Writing – review & editing, Supervision, Project administration, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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